

a What each model ran in Galaxy

Galaxy runs with at least one completed job in each family (%); UDTs were not offered on IWC

|                    |                                             | BixBench |     |      |          | CompBio |     |      |          | IWC     |     |      |          |
|--------------------|---------------------------------------------|----------|-----|------|----------|---------|-----|------|----------|---------|-----|------|----------|
|                    |                                             | GPT-5.5  | Sol | Luna | DeepSeek | GPT-5.5 | Sol | Luna | DeepSeek | GPT-5.5 | Sol | Luna | DeepSeek |
| Data handling      | Tables and text                             | 10       | 38  | 51   | 33       | 17      | 28  | 43   | 33       | 0       | 30  | 30   | 33       |
|                    | Format conversion and upload                | 19       | 42  | 57   | 23       | 10      | 16  | 31   | 22       | 10      | 27  | 23   | 10       |
|                    | Inspection and quality control              | 2        | 2   | 2    | 0        | 10      | 22  | 26   | 21       | 0       | 17  | 20   | 10       |
|                    | Data retrieval                              | 0        | 0   | 0    | 0        | 0       | 2   | 6    | 4        | 10      | 13  | 10   | 13       |
| Scientific methods | Read processing and alignment               | 2        | 2   | 2    | 2        | 8       | 16  | 18   | 15       | 27      | 40  | 33   | 30       |
|                    | Variant calling and annotation              | 0        | 0   | 0    | 0        | 3       | 6   | 10   | 7        | 0       | 0   | 0    | 3        |
|                    | Expression and differential testing         | 3        | 6   | 9    | 5        | 2       | 3   | 4    | 5        | 20      | 17  | 20   | 10       |
|                    | Single-cell and spatial analysis            | 0        | 0   | 0    | 0        | 1       | 4   | 5    | 7        | 3       | 3   | 0    | 10       |
|                    | Genomic intervals and sequence features     | 2        | 2   | 2    | 2        | 8       | 15  | 26   | 23       | 3       | 10  | 7    | 13       |
|                    | Phylogenetics                               | 12       | 16  | 21   | 17       | 0       | 0   | 0    | 0        | 0       | 0   | 0    | 0        |
|                    | Sequence search, taxonomy and assembly      | 0        | 1   | 1    | 1        | 4       | 5   | 8    | 7        | 10      | 27  | 27   | 10       |
|                    | Peak calling and epigenomics                | 0        | 0   | 0    | 0        | 1       | 1   | 2    | 2        | 3       | 10  | 10   | 3        |
|                    | Statistics, machine learning and enrichment | 22       | 35  | 34   | 19       | 2       | 4   | 9    | 9        | 0       | 0   | 0    | 0        |
|                    | Other methods                               | 0        | 0   | 0    | 0        | 0       | 0   | 0    | 0        | 3       | 10  | 7    | 10       |
|                    | UDT (agent-written code)                    | 66       | 35  | 39   | 9        | 56      | 75  | 41   | 35       | 0       | 0   | 0    | 0        |